
****4. `engine/src/hollow.rs`****

``rust

/*

* Adaptive Hollow Reflector - Endpoint Edition
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*/

use serde::{Deserialize, Serialize};

#[derive(Debug, Clone, Serialize, Deserialize)]

pub enum UserPrivilege {

Standard,
Admin,
DomainAdmin,

}

#[derive(Debug, Clone, Serialize, Deserialize)]

pub enum EndpointAction {

LogOnly,
SuspendProc,
KillTree,
IsolateHost,
SnapshotLock,
Sinkhole,
RevokeSession,
TriggerMfaReprompt,
SendSocWebhook,
FlagForHumanReview,

}

#[derive(Debug, Clone, Serialize, Deserialize)]

pub struct FileHollow {

// Core AHR fields

pub proc_hash: String, // sha256(exe_path + cmdline + parent_hash)
pub entropy_delta: f32, // avg entropy of last 50 writes
pub write_velocity: u32, // files/sec
pub rename_burst: bool, // mass .docx -> .lockbit
pub backup_tamper: bool, // vssadmin, wmic shadowcopy delete
pub c2_contact: bool, // outbound to .onion, mega.nz
pub decoy_touched: bool, // hit honeypot dir

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pub risk_score: u8,          // 1-10
pub last_seen: u64,         // unix timestamp
pub is_decoy: bool,

// META ADDITIONS by Gregory Scott Davis
pub peer_risk: f32,         // Jaccard similarity to LockBit, BlackCat, REvil
pub propagation_score: u32, // # adjacent endpoints with same proc_hash in 60s
pub mitre_ttp: Vec<String>, // T1486 (Encrypt), T1490 (VSS), T1048 (Exfil)
pub threat_intel_match: bool, // VirusTotal/Meta internal hash DB hit
pub user_privilege: UserPrivilege,
pub geo_fencing_risk: bool, // Login from anomalous ASN during business hours
pub clipboard_monitor: bool, // Detected crypto wallet address replacement
}

```

```

impl FileHollow {
  pub fn new(proc_hash: String) -> Self {
    Self {
      proc_hash,
      entropy_delta: 0.0,
      write_velocity: 0,
      rename_burst: false,
      backup_tamper: false,
      c2_contact: false,
      decoy_touched: false,
      risk_score: 1,
      last_seen: 0,
      is_decoy: false,
      peer_risk: 0.0,
      propagation_score: 0,
      mitre_ttp: Vec::new(),
      threat_intel_match: false,
      user_privilege: UserPrivilege::Standard,
      geo_fencing_risk: false,
      clipboard_monitor: false,
    }
  }
}

```